

Distribution of rupture sites and blebs on intracranial aneurysm walls suggests distinct rupture patterns in ACom and MCA aneurysms

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Abstract

The mechanisms behind intracranial aneurysm formation and rupture are not fully understood, with factors such as location, patient demographics, and hemodynamics playing a role. Additionally, the significance of anatomical features like blebs in ruptures is debated. This highlights the necessity for comprehensive research that combines patient-specific risk factors with a detailed analysis of local hemodynamic characteristics at bleb and rupture sites. Our study analyzed 359 intracranial aneurysms from 268 patients, reconstructing patient-specific models for hemodynamic simulations based on 3D rotational angiographic images and intraoperative videos. We identified aneurysm subregions and delineated rupture sites, characterizing blebs and their regional overlap, employing statistical comparisons across demographics, and other risk factors. This work identifies patterns in aneurysm rupture sites, predominantly at the dome, with variations across patient demographics. Hypertensive and anterior communicating artery (ACom) aneurysms showed specific rupture patterns and bleb associations, indicating two pathways: high-flow in ACom with thin blebs at impingement sites and low-flow, oscillatory conditions in middle cerebral artery (MCA) aneurysms fostering thick blebs. Bleb characteristics varied with gender, age, and smoking, linking rupture risks to hemodynamic factors and patient profiles. These insights enhance understanding of the hemodynamic mechanisms leading to rupture events. This analysis elucidates the role of localized hemodynamics in intracranial aneurysm rupture, challenging the emphasis on location by revealing how flow variations influence stability and risk. We identify two pathways to wall failure—high-flow and low-flow conditions—highlighting the complexity of aneurysm behavior. Additionally, this research advances our knowledge of how inherent patient-specific characteristics impact these processes, which need further investigation.

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Acom, bleb, hemodynamics, intracranial aneurysm, MCA, rupture

1 | INTRODUCTION

The formation and rupture of intracranial aneurysms are not fully understood but are believed to stem from vascular wall degradation due to abnormal hemodynamics.¹ Factors such as aneurysm location, patient age, hypertension, and smoking history play a role in this degradation and have been used to assess rupture risk.^{2,3} Neurosurgeons rely mainly on these factors for treatment decisions, although their underlying mechanisms are unclear. Intracranial aneurysms pose a significant life threat, making accurate rupture assessment based on a detailed understanding of the underlying mechanisms leading to rupture highly valuable.

Studies have shown that the rupture risk depends on the aneurysm location. Specifically, it is higher in the anterior communicating artery (ACoM), followed by the posterior communicating artery (PCoM), middle cerebral artery (MCA), and internal carotid artery (ICA). However, the underlying reasons for this link between aneurysm location and rupture risk remain unclear.^{4,5}

Women are generally at higher risk for harboring aneurysms, a phenomenon not entirely accounted for by typical rupture risk factors. Key contributors that have been considered include hormonal changes at menopause, anatomical differences, and a greater impact of smoking. On the other hand, studies have shown higher rupture rates in men, particularly in MCA bifurcation aneurysms. Although not entirely clear, this may be due to differences in arterial bifurcation and hemodynamic stresses between the sexes.^{6,7}

Although the underlying causes of intracranial aneurysm rupture are not fully understood, it seems that ruptures tend to affect a younger demographic compared to other cardiovascular diseases, with a median age of 60 years at the time of rupture.⁸ Clinical rupture risk factors also include hypertension and smoking history. Current and former smokers are more likely to experience ruptured aneurysms. Hypertension increases the risk and contributes to tissue remodeling and aneurysmal wall destabilization.^{9–11}

Computational fluid dynamics (CFD) has been used to analyze hemodynamic factors like wall shear stress (WSS), oscillatory shear index (OSI), and various flow patterns, which are thought to be key determinants of aneurysm rupture.¹² However, most studies have compared ruptured to unruptured aneurysms without identifying the exact rupture site, potentially introducing bias. Furthermore, the actual rupture often occurs at a single point, making whole-aneurysm comparisons less meaningful. The hemodynamics at the rupture site have been analyzed in very limited cases, with roughly 58 rupture sites from 13 published studies.^{13–25} However, the results are confusing, particularly the role of WSS.

On the one hand, Cebal et al.¹⁶ and Hejcl et al.²¹ have suggested that high WSS at the rupture point is a common feature. This is further supported by Hodis et al.,¹⁵ who also found elevated WSS and pressure near the rupture site. On the other hand, studies by Fukazawa et al.¹⁷ and Omodaka et al.¹³ have identified low WSS at the rupture points. These findings of low WSS have been supported by other studies.^{14,18–20,22–25} High OSI at the rupture point consistently appears as a parameter of importance across many studies.^{13,19,23} Additionally, both Kono et al.¹⁴ and Hodis et al.¹⁵ identified the rupture site as coinciding with an “impingement zone” during peak systole. These apparent disagreements could be since different studies included aneurysms at different locations. The hemodynamics, including WSS and OSI, can be significantly different for aneurysms at different locations. As such, studies that predominantly include MCA aneurysms (45 out of 58) may not be generalizable to aneurysms of other locations.

The role of blebs in aneurysm rupture is another area of research. Of the 13 studies analyzing rupture sites, only 8 identify blebs at rupture sites,^{13,14,16,18,19,22,23,25} while three do not address the issue despite the clear presence of blebs at the aneurysm's rupture locations.^{17,20,24} Moreover, despite the prevailing notion that blebs are common rupture sites, only seven out of 58 identified rupture points in the literature are explicitly associated with blebs. None of the studies delve into the specific characteristics of the blebs, such as whether they are thin-walled or thick, that may make them prone to rupture.

The aim of this research is multi-faceted. First, we conducted a detailed analysis of the distribution of 31 observed rupture sites, 34 thin blebs, and 28 thick blebs across different subregions of the aneurysm. Secondly, as observed in our previous study, we investigate the possible associations between rupture sites and blebs and connect these phenotypes to local hemodynamic characteristics.²⁶ Furthermore, this analysis is stratified by patient demographics and risk factors such as age, sex, aneurysm location, hypertension status, and smoking status. The ultimate goal is to improve our understanding of the mechanisms driving wall changes that eventually lead to wall failure and aneurysm rupture.

2 | METHODS

2.1 | Aneurysm datasets

This study analyzes data from 359 aneurysms in 268 patients who underwent surgical clipping. We utilized datasets that included pre-operative 3D rotational angiography (3DRA) and computed tomography angiography (CTA) images, patient demographics (sex, age, smoking, and hypertension history), and intraoperative videos from the surgical procedures. However, videos were available for only 130 aneurysms. Building on our prior work,²⁶ we reconstructed patient-specific vascular models from the 3DRA and CTA images and performed CFD simulations to assess hemodynamics. The data, summarized in Table 1, received approval from our Institutional Review Board (IRB).

2.2 | Rupture site identification and marking

In a subset of 31 ruptured aneurysms, the rupture site could be observed in the operative videos. Typically, in these videos, the rupture location is seen as either a hematoma attached to the wall at the rupture site or as a hole in the aneurysm wall. Once identified, the rupture site was interactively marked on the 3D vascular model using previously developed tools.²⁷ This step requires careful inspection of the operative videos and the 3D vascular model to understand the vascular anatomy and orientation of the aneurysm in the operative view and subsequently delineate (paint) the rupture site on the 3D model. Several examples are presented in Figure 1, where only one frame of the videos is shown for each case, along with the rupture site marked on the corresponding 3D vascular model.

2.3 | Bleb identification and marking

As in previous studies,²⁸ blebs were visually identified in the pre-operative 3D images and interactively marked on the reconstructed vascular models. Additionally, available intraoperative videos were inspected, and if the blebs were visible, they were labeled according to their color as “thin” (red) or “thick” (white or yellow). Two examples are shown in Figure 2: one thick hyperplastic (top row, white) and one thin (bottom row, red). A total of 213 blebs were identified in 359 aneurysms in our previous work.²⁶ Of these, 62 were visible in the intraoperative videos, 34 were labeled as thin (red), and 28 were labeled as thick (white or yellow).

TABLE 1 Patient and aneurysm characteristics.

Category	Value	Aneurysms	Blebs (incl. observable thin and thick)			Observed rupture sites
			All	Thin	Thick	
Sex	Males	64	31	6	8	9
	Females	206	108	25	19	18
Age (56 ± 10 , 25–81)	Young (<57)	139	65	17	12	19
	Old (>57)	130	73	14	15	8
Smoking history	No	78	42	13	8	11
	Yes	156	78	15	19	10
Hypertension	No	101	44	9	10	12
	Yes	133	80	22	17	10
Location	ACom	68	54	10	4	11
	PCom	40	26	4	4	2
	MCA	152	89	16	18	15
	ICA	66	26	1	2	2
Total		359	213	34*	28*	31*

Note: Aneurysm locations: ACom, PCom, MCA, and ICA. Bold values are observable thin and thick regions, and rupture sites.

Abbreviations: ACom, anterior communicating artery; ICA, internal carotid artery; MCA, middle cerebral artery; PCom, posterior communicating artery.

2.4 | Aneurysm subdivision

In our previous study,²⁶ the aneurysm wall was subdivided into three different sets of subregions: (a) “saccular subdivision”: neck, body, and dome subregions (based on the distance to the neck); (b) “streamwise subdivision”: inflow, central, and outflow subregions (based on distance along streamlines); and (c) “topographical subdivision”: bleb, parent aneurysm, and transition subregions (based on distance to bleb regions). Local hemodynamic characteristics, including WSS, WSS gradient (GRD), WSS divergence (DIV), OSI, and relative residence time (RRT), as well as critical points of the WSS vector field (i.e., saddle points, focus sources and sinks, and node sources and sinks), were then determined for each subregion. Examples of aneurysm subdivisions are presented in Figure 3.

2.5 | Analysis of rupture site and bleb location on aneurysm wall

First, the percentage of the area of each rupture site ($n = 31$) overlapping with the different saccular (neck, body, and dome), streamwise (inflow, central, and outflow), and topographical (bleb, transition, and parent) subregions of the aneurysm were calculated. These percentages were compared using Kruskal-Wallis analysis of variance tests for all rupture sites and across aneurysm anatomical locations, patient sex and age group, and history of hypertension and smoking. Pairwise comparisons between percentages of each category (e.g., female sex) in each subregion (e.g., neck)

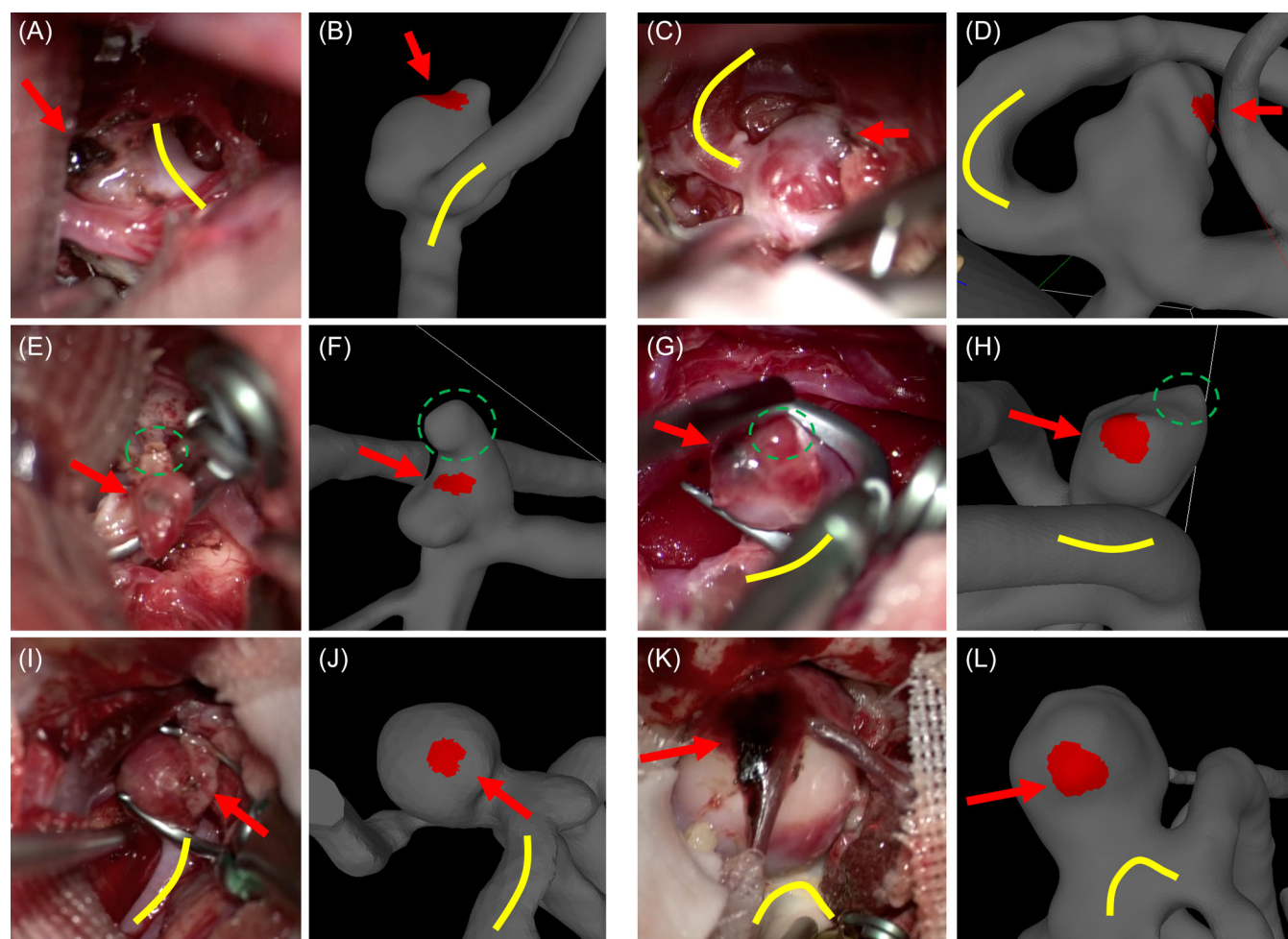


FIGURE 1 Identification of aneurysm rupture sites in intraoperative videos and marking on corresponding 3D vascular models. Red arrows point to the rupture sites in the videos and the marked region on the vascular models (also in red). Yellow lines identify arterial branches in the videos and models, and green dotted-line circles indicate aneurysm blebs. These anatomical features help us understand the orientation of aneurysms and vasculature in the videos.

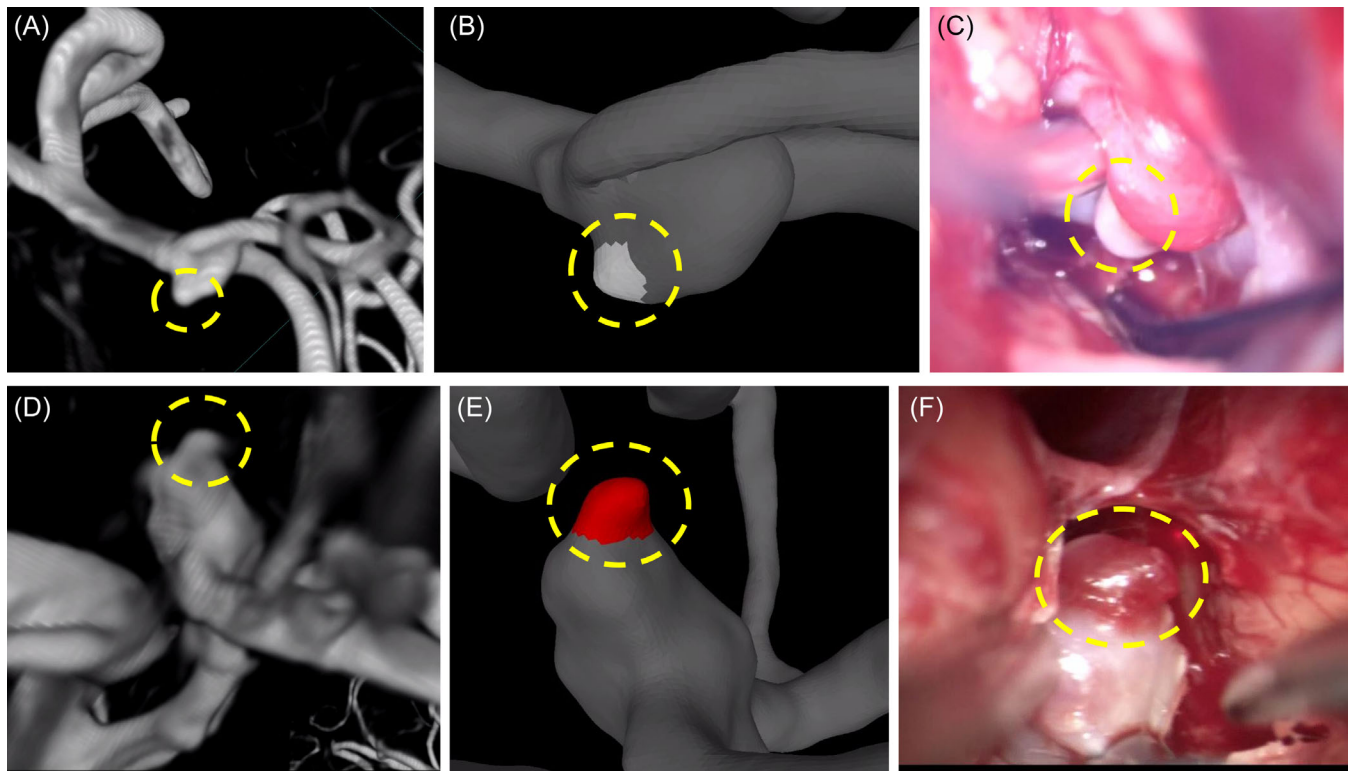


FIGURE 2 Identification, marking, and labeling of blebs. (A and D) Identification of blebs in pre-operative 3D rotational angiography (3DRA) images; (B and E) bleb marking on corresponding 3D vascular models; and (C and F) labeling of blebs according to their color in the intraoperative videos (C: white, F: red).

were carried out using the nonparametric Wilcoxon (Mann–Whitney) test. Differences were considered significant if $p < .05$ (i.e., 95% confidence).

Next, the percentage of the bleb area overlapping with the different saccular and streamwise regions of the aneurysm was calculated and similarly compared for $n = 34$ “thin” blebs and $n = 28$ “thick” blebs. Only ACom and MCA aneurysms were statistically compared for anatomical locations since the other locations contained too few rupture sites or blebs visible in the videos (see Table 1).

3 | RESULTS

First, the percentage area distribution of rupture sites over the saccular, streamwise, and topographical regions are presented in Figures 4–6, respectively. These figures show the mean (bars) and standard deviation (error bars) in each region for all rupture sites (top left), then subdividing by aneurysm location (ACom, MCA) (top center), sex (top right), age group (bottom left), hypertension (bottom center), and smoking (bottom right). Similarly, the percentage area distribution of thin blebs and thick blebs over the saccular and streamwise regions are presented in Figures 7 and 8, respectively. The legend shows Kruskal–Wallis test p -values for within-subregion (e.g., ACom-saccular, etc.) comparisons. Additionally, Mann–Whitney test p -values are provided below each subregion (X-axis) within patient or aneurysm characteristic (e.g., ACom-neck vs. MCA-neck).

3.1 | Saccular distribution of rupture sites

In general, rupture sites involved more frequently the aneurysm dome followed by the body and less frequently the neck of the aneurysms ($p < .0001$). These associations were consistent across aneurysm location, patient sex and age group, hypertension, and smoking history. ACom aneurysms had a slightly larger proportion of rupture sites involving

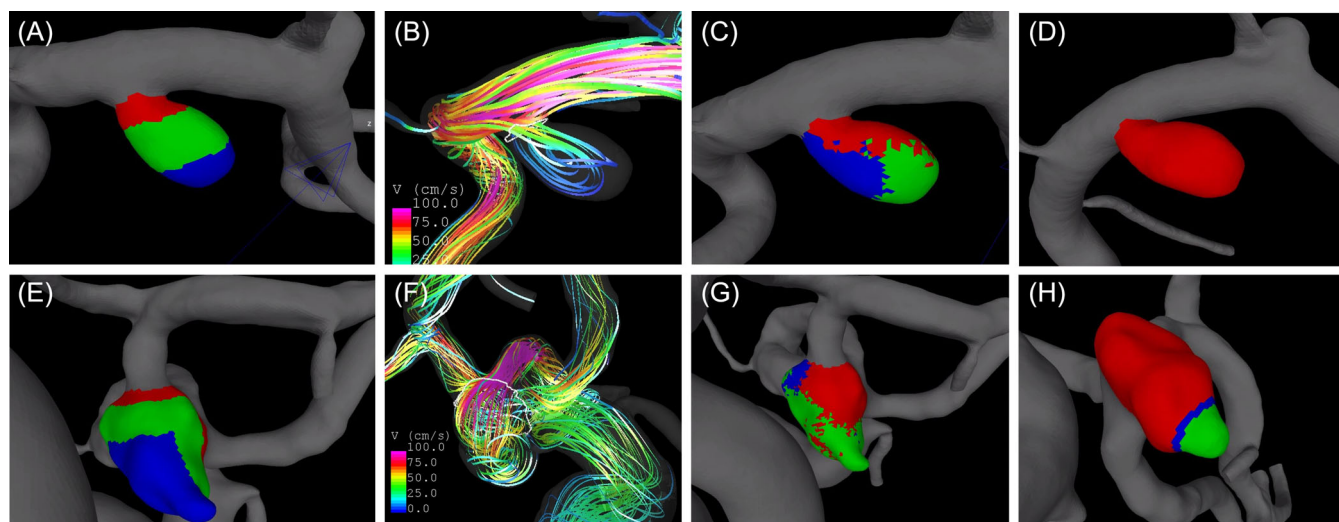


FIGURE 3 Examples of middle cerebral artery (MCA) (top row) and anterior communicating artery (ACoM) (bottom row) aneurysm subdivision: (A and E) saccular subdivision into neck (red), body (green), and dome (blue) regions; (B and F) flow streamlines; (C and G) streamwise subdivision into inflow (red), central (green), and outflow (blue) regions; and (D and H) topographical subdivision into bleb (green), transition (blue), and parent aneurysm (red) regions.

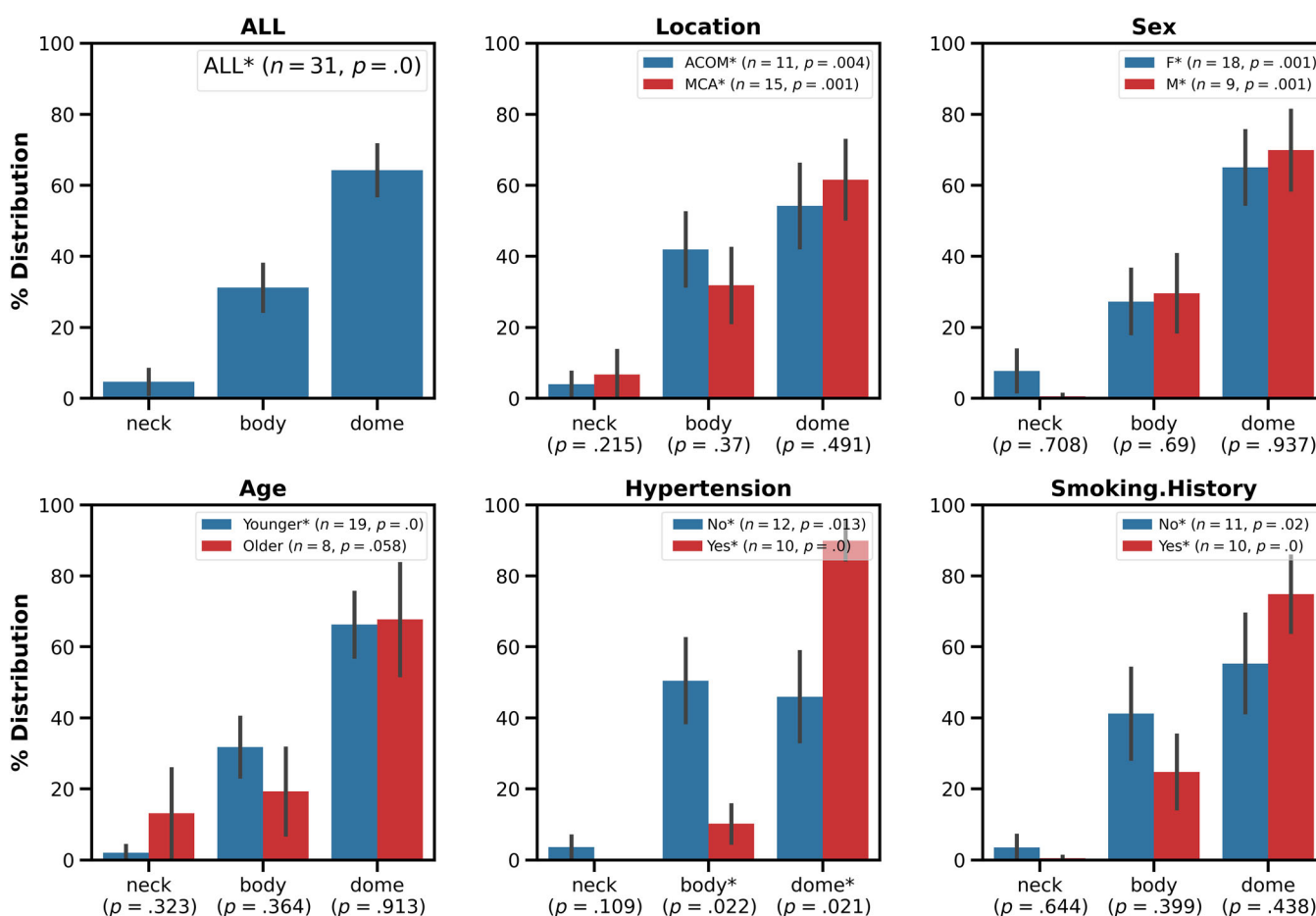


FIGURE 4 Saccular distribution of rupture sites.

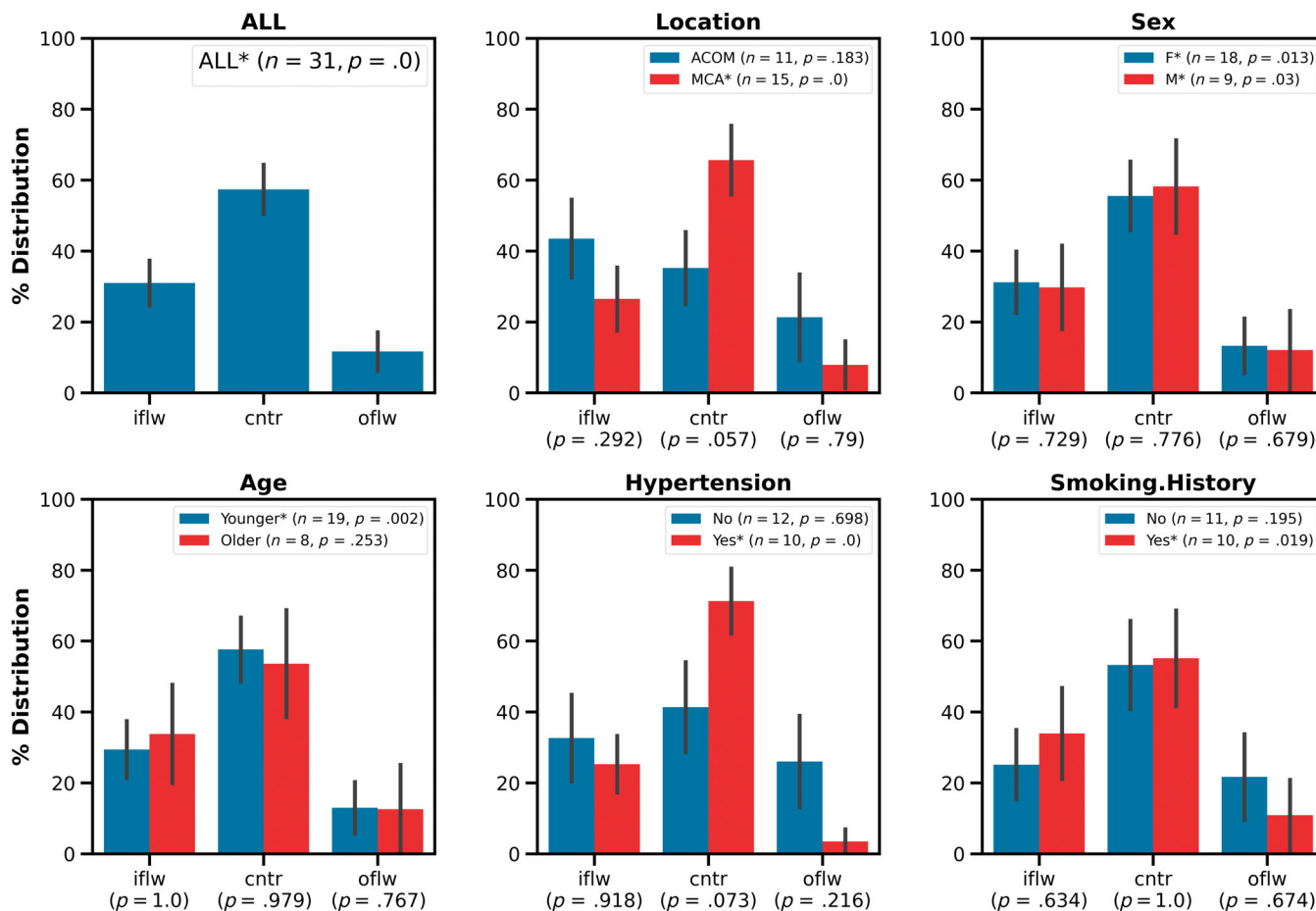


FIGURE 5 Streamwise distribution of rupture sites.

the body ($p = .37$) and fewer involving the dome ($p = .491$) than MCA aneurysms, but these trends did not reach statistical significance. No significant differences were observed between males and females and between younger and older patients. Interestingly, hypertensive patients had more rupture sites at the dome ($p = .021$) and fewer at the body ($p = .022$) (and none at the neck, $p = .109$) compared to normotensive patients. Patients with smoking history had slightly more ruptures involving the dome ($p = .438$) and fewer involving the body ($p = .399$) or neck ($p = .644$) compared to nonsmokers, but these trends did not reach statistical significance.

3.2 | Streamwise distribution of rupture sites

Rupture sites were generally more frequent in the central region, followed by the inflow and outflow regions ($p < .0001$). These associations were consistent across patient sex, age group, hypertension, and smoking history. However, compared to MCA aneurysms, which followed this general trend ($p < .0001$), this distribution was different for ACom aneurysms, where the most frequent region was the inflow, followed by the central and lastly, the outflow region with smaller but not statistically significant proportions ($p = .183$). Interestingly, hypertensive patients tended to have a larger proportion of rupture sites involving the central stream region ($p = .073$) compared to normotensive patients.

3.3 | Topographical distribution of rupture sites

Rupture sites were most frequently associated with aneurysm blebs or their transition regions and less frequently with the parent aneurysm ($p < .0001$). This association was largely consistent across age groups and smoking history. However, this distribution differed slightly for ACom ($p = .074$) and MCA ($p = .003$) aneurysms. MCA aneurysms tended

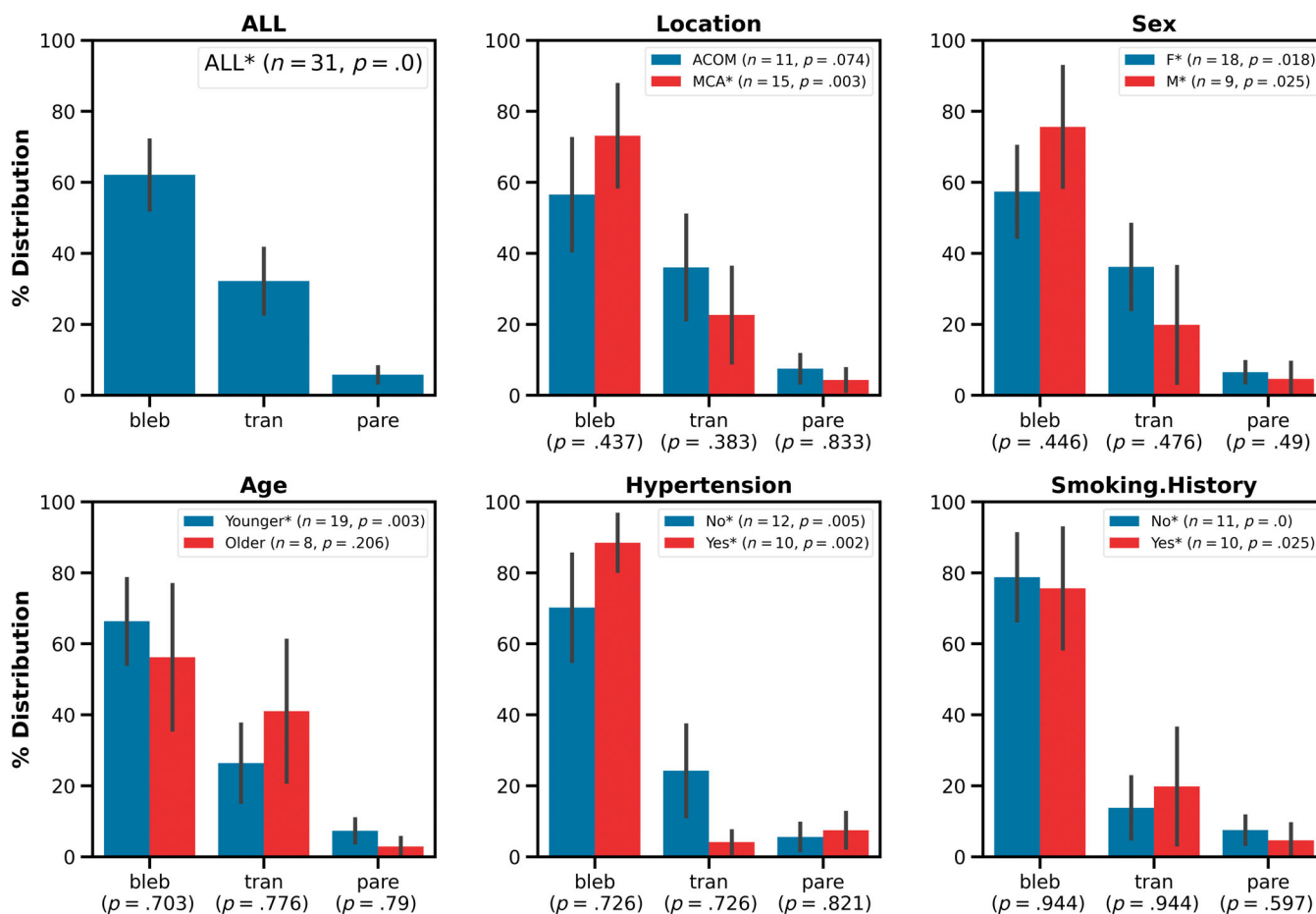


FIGURE 6 Topographical distribution of rupture sites.

to have more rupture sites associated with blebs ($p = .003$) than ACom aneurysms, which had a slightly larger proportion involving the transition region ($p = .074$). Additionally, hypertensive and male patients tended to have fewer rupture sites involving the transition region and slightly more at the blebs compared to normotensive and female patients, respectively. However, these differences did not reach statistical significance.

3.4 | Saccular distribution of blebs

Blebs are generally most commonly found at the aneurysm body, followed by the dome, and less frequently at the neck ($p > .0001$). Although thin and thick blebs follow this general trend, the number of thin blebs is slightly larger than the number of thick blebs at the neck ($p = .107$) and smaller at the dome ($p = .535$). However, these differences did not reach significance at 95% confidence. Interestingly, in ACom aneurysms, thin blebs were more frequent at the dome than at the body and neck ($p = .014$), and the number of thin blebs at the dome was marginally larger than those in MCA aneurysms ($p = .061$). In contrast, thick blebs followed roughly identical distributions in ACom and MCA aneurysms. The distributions of thin blebs were different in males, where the majority involved the dome followed by the body and the neck ($p = .015$), compared to females, where the distribution followed the overall trend ($p < .0001$). Additionally, compared to males, females had more thin blebs at the neck ($p = .113$) and a smaller number at the dome ($p = .017$). The distributions of thick blebs were also different between females (more frequent at the dome than body and neck, $p < .0001$) and males, which followed the overall trend ($p = .001$). Older patients had more thin blebs involving the aneurysm neck ($p = .037$), while younger patients had more thin blebs involving the dome ($p = .006$). The distributions of thick blebs were similar between older and younger patients. The thin and thick blebs distributions were not statistically different between hypertensive and normotensive patients. The distributions of thin blebs were not

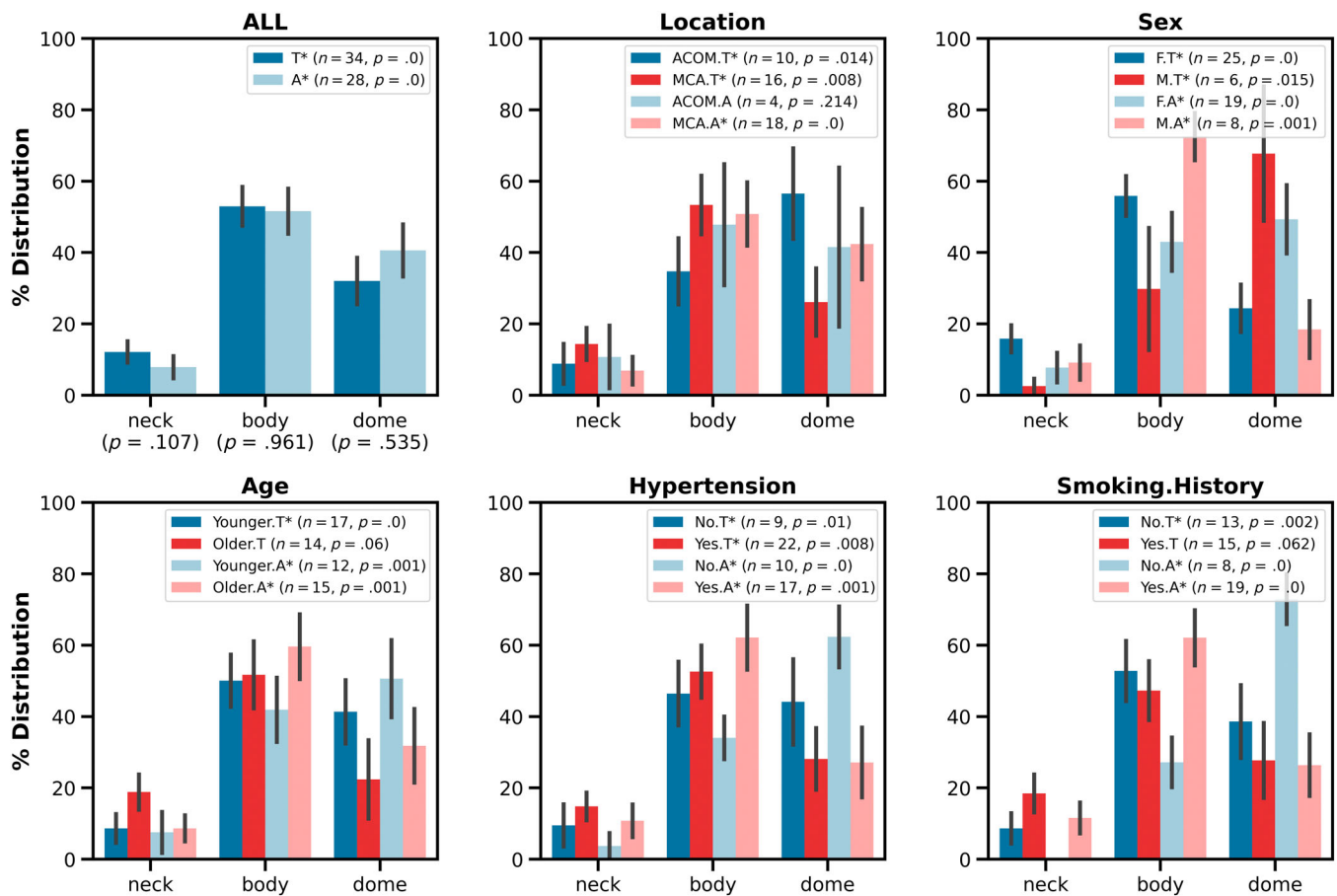


FIGURE 7 Saccular distribution of blebs.

different between patients with and without smoking history. However, patients with a smoking history had more thick blebs at the aneurysm body ($p = .018$), while patients without a smoking history had more at the dome ($p = .01$).

3.5 | Streamwise distribution of blebs

In general, blebs occurred most frequently in the central stream region, followed by the outflow region and subsequently in the inflow region ($p < .0001$). This association was consistent for both thin blebs and thick blebs. However, thin blebs were slightly more frequent in the inflow region ($p = .109$) and less frequent in the outflow region ($p = .132$) than thick blebs. Still, these differences did not reach statistical significance at 95% confidence. This distribution was also consistent across aneurysm locations (ACOM and MCA), hypertensive and normotensive patients, and patients with and without a smoking history. Females had a larger proportion of thin than thick blebs in the inflow region ($p = .038$). Younger patients had a larger number of thin blebs in the central region than older patients ($p = .002$).

4 | DISCUSSION

Despite the prevailing conviction that intracranial aneurysms rupture predominantly at the aneurysm dome in association with blebs, little quantitative data in the literature supports this notion. Furthermore, most studies do not differentiate between blebs with different wall characteristics, which presumably develop due to distinct local hemodynamic and biomechanical environments. The current study investigated the location of rupture sites and blebs on the aneurysm sac, how they are affected by aneurysm location and patient characteristics, and the hemodynamic conditions that generally prevail in different regions of the aneurysm, as studied previously.²⁶

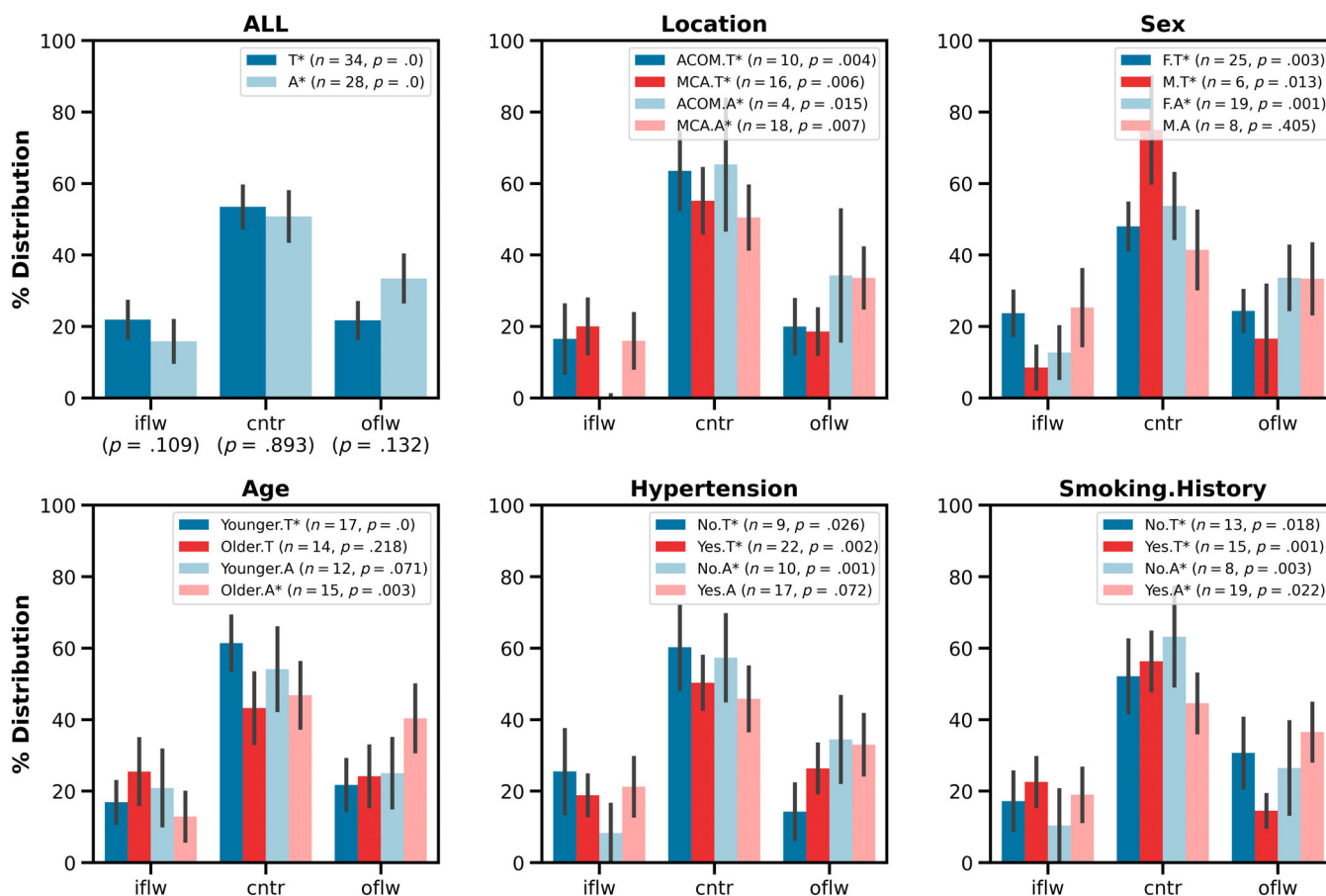


FIGURE 8 Streamwise distribution of blebs.

4.1 | Saccular aneurysm regions

The dome was the saccular region of the aneurysm that most frequently involved sites of wall failure (i.e., rupture sites). Although this finding was consistent across aneurysm locations, it was more prevalent among MCA aneurysms compared to ACom aneurysms, as well as in patients with hypertension or smoking history. Although the dome was a common location for bleb development, it was not the most frequent. In general, at the dome, there were more thick blebs than thin blebs. However, thin blebs were more frequent at the dome of ACom aneurysms in male and younger patients. In contrast, thick blebs were more frequent at the dome of females and nonsmokers. The dome typically experienced conditions of low flow, marked by low WSS, minimal WSS GRDs, and extended residence times, particularly in MCA and PCom aneurysms.²⁶

On the other hand, ACom aneurysms, as well as normotensive and non-smoker patients, had more rupture sites in the body compared to MCA aneurysms, hypertensive, or smoker patients. The body was also the most common location for bleb development, and smokers had thicker blebs on the body compared to non-smokers. The body of the aneurysm was subject to slower (greater RRT) and more oscillatory (higher OSI) flows, particularly in ICA aneurysms, and it held the highest count of focus and node critical points, such as vortex ends and flow impingements.²⁶

The neck was the least frequent saccular region involving rupture sites, and it was also a less frequent location for bleb development. However, there were more thin blebs at the neck than thick blebs and female and older patients had more thin blebs than male or younger patients. The neck, the saccular region of the aneurysm, typically encountered the highest WSS and WSS GRDs and had the greatest number of saddle points across aneurysms in all locations.²⁶ Additionally, the WSS DIV was positive and more significant (implying flow impingement) at the neck of ACom and MCA aneurysms, whereas it was negative (suggesting flow convergence) at the neck of ICA aneurysms.²⁶

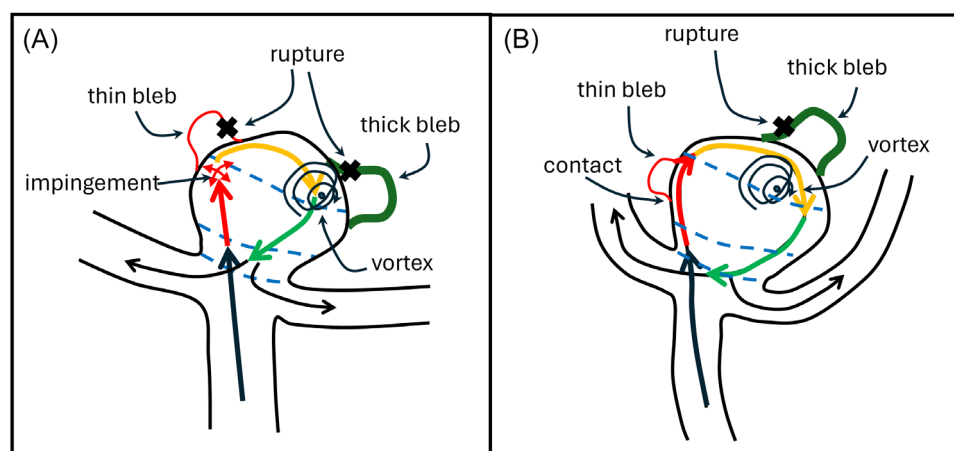


FIGURE 9 Summary of relationships between local flow conditions, bleb development, and rupture sites: (A) aneurysm with inflow jet impaction, high wall shear stress (WSS), and critical node point at the body where thin blebs are more common, and vortex flow with low and oscillatory WSS at the dome where thick blebs are more common and (B) aneurysm with inflow-wall contact with associated high WSS and WSS gradient where thin blebs are observed (but less frequently), and vortex flow with low and oscillatory WSS at the dome where thick blebs are more common. Rupture sites (marked with an “x”) are most commonly associated with blebs.

4.2 | Streamwise aneurysm regions

The inflow was the most frequent location of rupture sites in ACom aneurysms (although not the most frequent when combined with aneurysms at other locations). The inflow also harbored larger numbers of thin blebs compared to thick blebs, especially for females. The inflow region of the hemodynamic conditions was marked by the highest WSS and WSS GRDs and had the greatest number of saddle points.²⁶

The central flow region was, in general, the most frequent location for aneurysm rupture (except in ACom aneurysms), especially for hypertensive patients. It was also the most frequent location for bleb development, but thin blebs were less frequent, and younger patients had more thin blebs at the central flow region compared to older patients. The central region experienced slow flows, characterized by the smallest WSS and WSS GRDs, yet it featured the largest oscillations (OSI) and residence times (RRT). This region also held the highest number of focus and node critical points, typically the endpoints of vortex swirling flows.²⁶

Rupture sites were infrequently located in the outflow region, and blebs were relatively common but not the most frequent at the outflow. The outflow region exhibited the greatest (positive) WSS DIV in ACom, PCom, and MCA aneurysms, indicating significant secondary flow impingement.²⁶

4.3 | Topographical aneurysm regions

Blebs were the most frequent location of aneurysm rupture, especially in MCA aneurysms and hypertensive and male patients. Interestingly, in our sample, blebs were under higher WSS and WSS GRD compared to the transition and parent aneurysm regions.²⁶ They also held a large number of saddle points in ACom and PCom aneurysms.

The transition zone was a common location of rupture sites but not the most frequent. It was more frequent in ACom than in MCA. Similarly, rupture sites were more frequent at the transition zone in females compared to males and in normotensive compared to hypertensive patients. The transition zone was distinguished by the highest oscillatory flow (OSI) and had the highest number of saddle points in ICA aneurysms compared to other locations.²⁶

Finally, the parent aneurysm region was the least frequent location of rupture sites.

4.4 | Key findings

The findings of this study are schematically summarized in Figure 9. In ACom aneurysms (A), typical flows include inflow jet impingement with associated high WSS and WSS GRD and node critical points at the body of the aneurysm,

TABLE 2 Location of rupture sites in previous studies.

Ref. studies (nr RS, nr visible)	Aneurysm location	Observed RS location in An. region			Hemodynamics at RS
		Saccular	Streamwise	Topographical	
Omodaka et al. ¹³ (<i>n</i> = 6, vis = 6)	MCA	Body	Inflow	Bleb	Low WSS + high OSI
	MCA	Dome	Central	Parent an.	
	MCA x4	Dome	NA	NA	
Fukazawa et al. ¹⁷ (<i>n</i> = 12, vis = 4)	MCA	Neck	Outflow	Bleb	Low WSS + low flow (complex flow structures) WSS: RS $\ll$ dome
	MCA	Dome	Inflow	Bleb	
	MCA	Dome	Central	Bleb	
	MCA	Dome	Central	Parent an.	
	MCA x8	NA	NA	NA	
Janiga et al. ¹⁸ (<i>n</i> = 1, vis = 1)	MCA	Dome	Inflow	Bleb	Low WSS + high OSI
Sejkorová et al. ¹⁹ (<i>n</i> = 1, vis = 1)	MCA	Body	Inflow	Bleb	Low WSS
Wang et al. ²² (<i>n</i> = 1, vis = 1)	MCA	Dome	Central	Bleb	Low WSS + high OSI
Suzuki et al. ²⁴ (<i>n</i> = 7, vis = 5)	MCA	Dome	Central	Bleb	Low WSS + high Pressure
	MCA	Body	Central	Parent an.	
	MCA	Dome	Central	Parent an.	
	MCA x2	Dome	Central	Transition	
	MCA x2	NA	NA	NA	
Jiang et al. ²⁵ (<i>n</i> = 1, vis = 1)	MCA	Dome	Central	Bleb	Low WSS, WSSG + high OSI and pressure and GON + impingement jet.
Cebal et al. ¹⁶ (<i>n</i> = 9, vis = 9)	ICA	Dome	Central	Parent an.	Abnormally high WSS (8) and abnormally low WSS (1 PCom)
	MCA x2	Dome	Central	Parent an.	
	PCom	Dome	Central	Transition	
	PCom x4	Dome	Central	Parent an.	
	ACom	Body	Inflow	Parent an.	
Hodis et al. ¹⁵ (<i>n</i> = 1, vis = 1)	ACom	Body	Inflow	Parent an.	High WSS + high pressure + impingement jet
Hejčl et al. ²¹ (<i>n</i> = 1, vis = 1)	ACom	Dome	Central	Parent an.	High WSS + high OSI + high pressure + impingement jet
Kono et al. ¹⁴ (<i>n</i> = 1, vis = 1)	ACA	Dome	Central	Bleb	Low WSS at end diastole + high pressure at peak systole at impingement
Zhang et al. ²⁰ (<i>n</i> = 1, vis = 1)	PCom	Dome	Inflow	Bleb	Low WSS + high OSI
Li et al. ²³ (<i>n</i> = 16, vis = 0)	ICA	NA	NA	NA	Low WSS + high OSI
	ACom				
	MCA x14				

Note: NA, rupture site not visible.

Abbreviations: OSI, oscillatory shear index; RS, rupture sites; WSS, wall shear stress.

where thin blebs are more frequently found. These aneurysms also typically feature vortex flows with low WSS and high OSI at the dome or body of the aneurysm, where thick blebs are more commonly observed. Rupture is common at both thin and thick blebs but seems to depend on the patient's smoking and hypertension. In MCA aneurysms (B), typical flows include an inflow jet that contacts and slides along the wall at the neck/body of the aneurysm and produces a region of high WSS and WSS GRD (although smaller than those associated with impingement), which favors thin bleb

development but less frequently than in AComs. Vortex flows with corelines moving around the body or dome of the aneurysm during the cardiac cycle, creating regions of low and oscillatory WSS, which are quite common in MCA aneurysms. These flows seem to induce thick bleb development in these regions of the aneurysm and are also favored in smoking and hypertensive patients.

4.5 | Possible implications

In ACom aneurysms, rupture sites seem to be related to “high” flow effects since they are more frequent at the inflow zone and the body of the aneurysm, where WSS and WSS GRDs are larger (and contain more saddle points) than at the dome or central flow zones. Furthermore, ACom aneurysms harbor more thin-red blebs in these regions than MCA aneurysms. Additionally, this is more prevalent in normotensive, non-smoker, and male patients.

In contrast, in MCA aneurysms, rupture sites are more frequently located at the dome and central flow region, which are exposed to “low” flow conditions characterized by lower and more oscillatory WSS and influenced by swirling vortical flows. These regions also contain a larger proportion of thick blebs. Additionally, the localization of rupture sites at the dome of MCA aneurysms is affected (increased) in patients with hypertension or smoking history as well as female patients.

This suggests two possible mechanisms of wall degeneration and ultimate failure: (a) development of thin-red blebs by decellularization and extra cellular matrix degradation processes in high flow areas associated with the aneurysm inflow jet, especially in ACom aneurysms, male patients without hypertension or smoking history; and (b) development of thick blebs by hyperplastic, remodeling, and inflammatory processes in low, oscillatory and vortex flow regions, especially in MCA aneurysms, females, and patients with hypertension or smoking history. Naturally, both conditions can be present simultaneously in the same aneurysm, resulting in heterogeneous walls that could be prone to rupture at different locations due to different mechanisms.

Certainly, the failure of the wall is also strongly influenced by the intramural stresses, which in turn are affected by the local wall curvature and thickness and the internal structure of the wall (i.e., collagen architecture), both of which are altered by the presence of a bleb. However, these effects were not directly investigated in the current study.

4.6 | Comparison with previous studies

Several previous studies have analyzed the local hemodynamics at rupture sites, either identified during open surgery (or intraoperative videos)^{13,17,21–24} or by analyzing blood extravasation in 3D images.^{14–16,18–20,25} Although these studies did not analyze the location of the rupture site with respect to different aneurysm regions, it was possible to determine the rupture site location with respect to the different aneurysm regions considered here of some aneurysms that were visible in the figures presented in these papers. These locations, along with the corresponding aneurysm location and hemodynamic conditions at the rupture sites reported in the studies, are summarized in Table 2.

The location of a total of 21 visible MCA aneurysms were assessed. Of these, 16 (76%) were located at the dome, 3 (14%) at the body, and only 1 (5%) at the neck of the aneurysms. The streamwise and topographical location of 17 of the 21 rupture sites could be assessed from the available figures. Of these, 12 (71%) were at the central flow region, 4 (24%) were at the inflow, and 1 (6%) was at the outflow. Likewise, 9 (53%) coincided with blebs, while 6 (35%) were in the parent aneurysm region and 2 (12%) in the transition region. In general, low flow conditions (low WSS and high OSI) were reported for these rupture sites.

These distributions of rupture sites with respect to saccular and streamwise regions agree with the corresponding distributions in MCA aneurysms in our study (Figures 4 and 5), but in previous studies, a larger proportion of MCA rupture sites were located in the parent aneurysm (or aneurysms without blebs).

Only three rupture sites reported in ACom aneurysms could be assessed. Of these, 2 (67%) were located at the body and inflow region, and 1 (33%) was at the dome and central flow regions. All 3 (100%) sites were in the parent aneurysm (no blebs in these cases). The studies consistently reported high flow conditions (high WSS and pressure) associated with flow impingement in all three ACom cases.

As in our study, more rupture sites in ACom aneurysms were located at the aneurysm body than in MCA aneurysms. The distribution of rupture sites with respect to streamwise regions in ACom aneurysms was similar to the distribution in our study (more rupture sites at the inflow, followed by the central and finally outflow regions), and different from the corresponding distribution in MCA aneurysms. Similarly, the majority of rupture sites were associated with blebs.

Thus, in general, our findings (including rupture site locations and prevalent hemodynamic conditions at the rupture sites) are in good agreement with the data from previous studies. These observations support the hypothesis that ACom and MCA aneurysms may follow different pathways to rupture (which could explain their different rupture rates), one based on wall degeneration and weakening in response to abnormally high flow conditions (more prevalent in ACom aneurysms) and another based on progressive wall remodeling and degradation, presumably mediated by inflammation, in response to abnormally low flow conditions (most prevalent in MCA aneurysms).

4.7 | CFD in IA management

CFD has emerged as a critical tool for understanding the complex interplay between hemodynamics and biological responses that lead to aneurysm progression and rupture. By analyzing flow characteristics such as vortices and impingement, CFD complements traditional evaluations based on geometry, anatomical location, and patient demographics. This integration extends to various imaging modalities, including 2D and 4D DSA, as well as 4D phase-contrast MR, enhancing our ability to assess aneurysms, particularly those that are small and regularly shaped yet rupture at significant rates—about 20% for aneurysms smaller than 10 mm without blebs.²⁹ Our study advances the precision of aneurysm management by providing insights into the local hemodynamic conditions and their impact on wall characteristics, offering a pathway toward more personalized clinical strategies. This combined approach not only elucidates the hemodynamic factors contributing to aneurysm behavior but also aids in identifying those atypical cases that, despite appearing less risky, still present a considerable risk of rupture.

4.8 | Limitations

The current study has several limitations. First, the CFD models are based on several approximations and assumptions. However, they yield reasonable results, as discussed in our previous work.^{26,27,30} Secondly, splitting the aneurysm into different subregions (saccular, streamwise, and topographical) is arbitrary but well-defined and reproducible. Third, identifying and marking rupture sites and blebs is subjective, and the sample size is small ($n = 31$ rupture sites, $n = 62$ blebs visible in videos). Similarly, the sample sizes become even smaller when split by different factors such as aneurysm location, patient sex, age, smoking, and hypertension. Fourth, other characteristics, such as aneurysm surface curvature, intramural stress, wall structure, etc., are not studied but could be important factors influencing the location of the wall failure and aneurysm rupture. Further studies should consider these factors and larger sample sizes. Despite these limitations, the current study contributes valuable data regarding the location of rupture sites and different kinds of blebs and their relationship to local hemodynamic conditions typically found in different subregions of the aneurysm. This information is important to understand further the mechanisms leading to aneurysm rupture.

5 | CONCLUSIONS

Several factors contribute to wall failure and aneurysm rupture, primarily through the development of blebs, which exhibit considerable variability in structure and composition. Our findings underscore that not all blebs are the same; some are thin blebs and decellularized, often influenced by high flow dynamics, whereas thick blebs may develop under conditions of low, oscillatory flow, possibly due to collagen deposition or atherosclerotic changes. The occurrence of ruptures is complex, manifesting across various pathological conditions within different bleb types, and sometimes away from blebs. This complexity is further compounded by factors such as aneurysm location, which determines local hemodynamic environments, and patient demographics like sex and age, which are linked to the prevalence of thin blebs. Additionally, lifestyle factors such as smoking and hypertension play a critical role in the formation of thick blebs and their propensity for rupture, particularly at the dome. This study enriches our understanding of the multifaceted

pathways leading to aneurysm wall degeneration, highlighting the interplay between non-modifiable patient characteristics and modifiable behavioral habits. Ultimately, this study enhances our ability to predict and manage aneurysm rupture by elucidating the diverse hemodynamic and biomechanical pathways leading to wall thinning (decellularization) or thickening (remodeling), thereby paving the way for more precise and personalized intervention strategies that focus on specific conditions conducive to rupture.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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